Internet Programming (sessional) [CSE-326]

Front End Report on:

CUET Medical Center

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1 Objectives

- To establish a centralized system for storing and managing patient,doctor and faculty-related data.
- To simplify the process of scheduling appointments, managing prescriptions, and beds.
- To enable real-time monitoring of bed availability and drug inventory to optimize resource utilization.
- To provide an AI based feature to the patient to obtain the solution of common diseases.

2 Introduction

In this project, we have used HTML, CSS, JavaScript to make the website look good.

HTML (Hypertext Markup Language)

HTML, or Hypertext Markup Language, is the main language used to build websites. It helps organize and display text, images, links, and other content on web pages so that browsers can show them properly. Think of it as the skeleton of a website that holds everything together in a clear and structured way.

CSS (Cascading Style Sheets)

CSS, which stands for Cascading Style Sheets, is the language used to style and design web pages. While HTML creates the structure of a website, CSS controls how everything looks, such as colors, fonts, layouts, and spacing. It helps make websites more attractive and easier to use by giving web pages a polished and consistent appearance.SS

JavaScript (JS)

JavaScript is a programming language that adds interactivity and dynamic behavior to websites. It lets you create things like buttons that respond when clicked, animations, forms that check your input, and much more. Basically, JavaScript makes websites come alive and allows users to interact with them in real time.

3 Front End Design

3.1 Use Case Diagram

Use case Diagram Description

The use case diagram illustrates the interactions between users and the system. Delineates the functionalities accessible to both regular users and administrations.

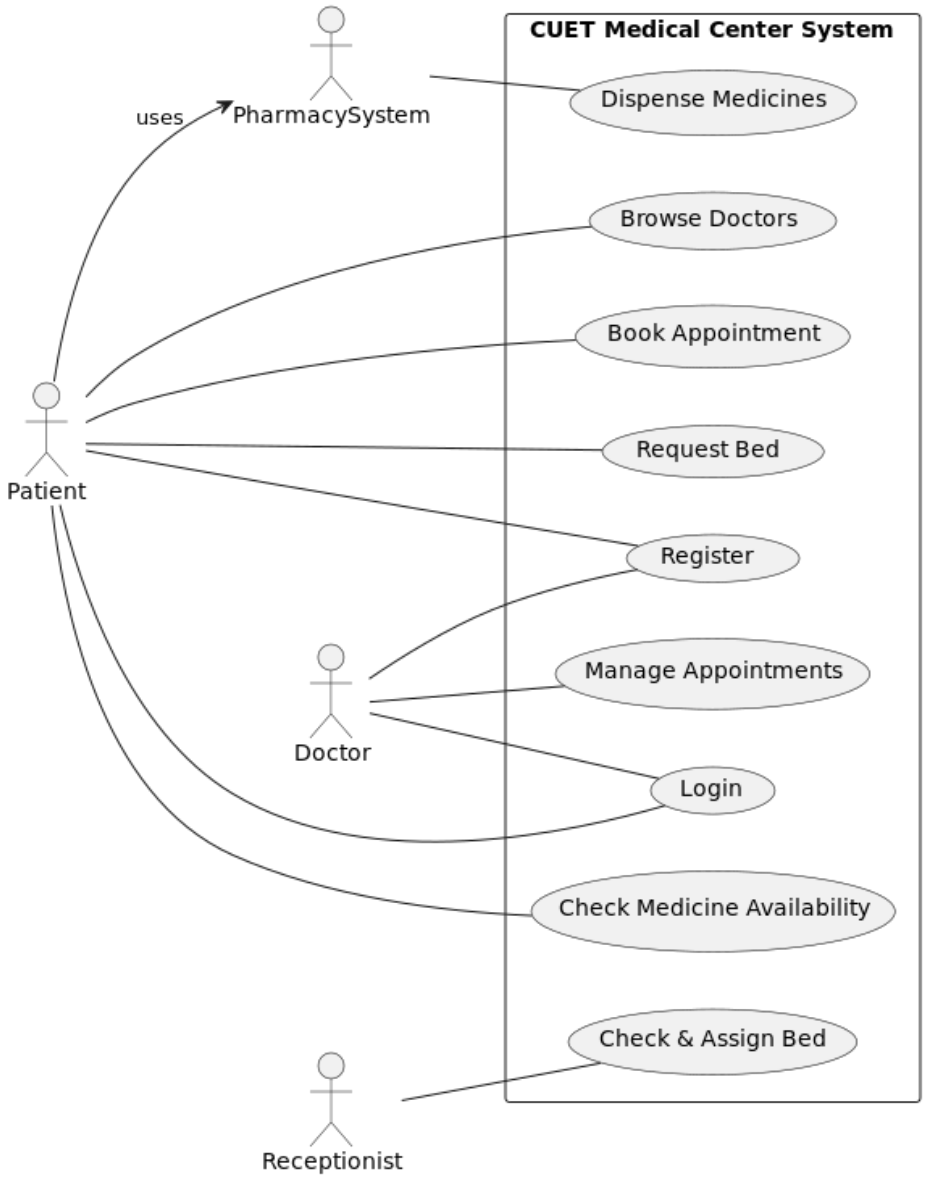

Figure 1: Use Case Diagram of CUET Medical Center

Actors

Patient Use Cases

Search: Patients can search for doctors, available services, and appointment slots within the system.
Book Appointment: Patients can schedule appointments with doctors according to their availability.
View Medical Records: Patients can access and review their medical history and prescriptions.
Request Bed: Patients can request bed allocation when admitted to the medical center.
Check Medicine Availability: Patients can check the availability of prescribed medicines in the pharmacy.

Doctor Use Cases

Manage Appointments: Doctors can view and manage their scheduled patient appointments.
Update Medical Records: Doctors can update patient medical records and treatment plans.
Prescribe Medicines: Doctors can prescribe medicines to patients based on diagnosis.
Login: Doctors can securely login to access the system and patient data.

Staffs Use Cases

Assign Beds: Staffs can check and assign beds to patients based on availability.
Manage Inventory: Staffs manage medicine stock and update inventory levels.
Dispense Medicines: Staffs dispense medicines to patients according to prescriptions of the doctor.
Login: Staffs can securely login to perform administrative tasks.

3.2 Activity Diagram:

Activity Diagram Description

- **Landing Page:**
  - Users choose to register or log in.
- **Patient:**
  - Browse doctors and book appointments.
  - Request beds and check availability of medicine/lab .
  - View medical records and logout.
- **Doctor:**
  - Manage appointments and update medical records.
  - Prescribe medicines and logout.
- **Staff:**
  - Check and assign beds.
  - Manage the inventory of medicine and dispense medicines.
  - Handle lab tests and notify of unavailability.

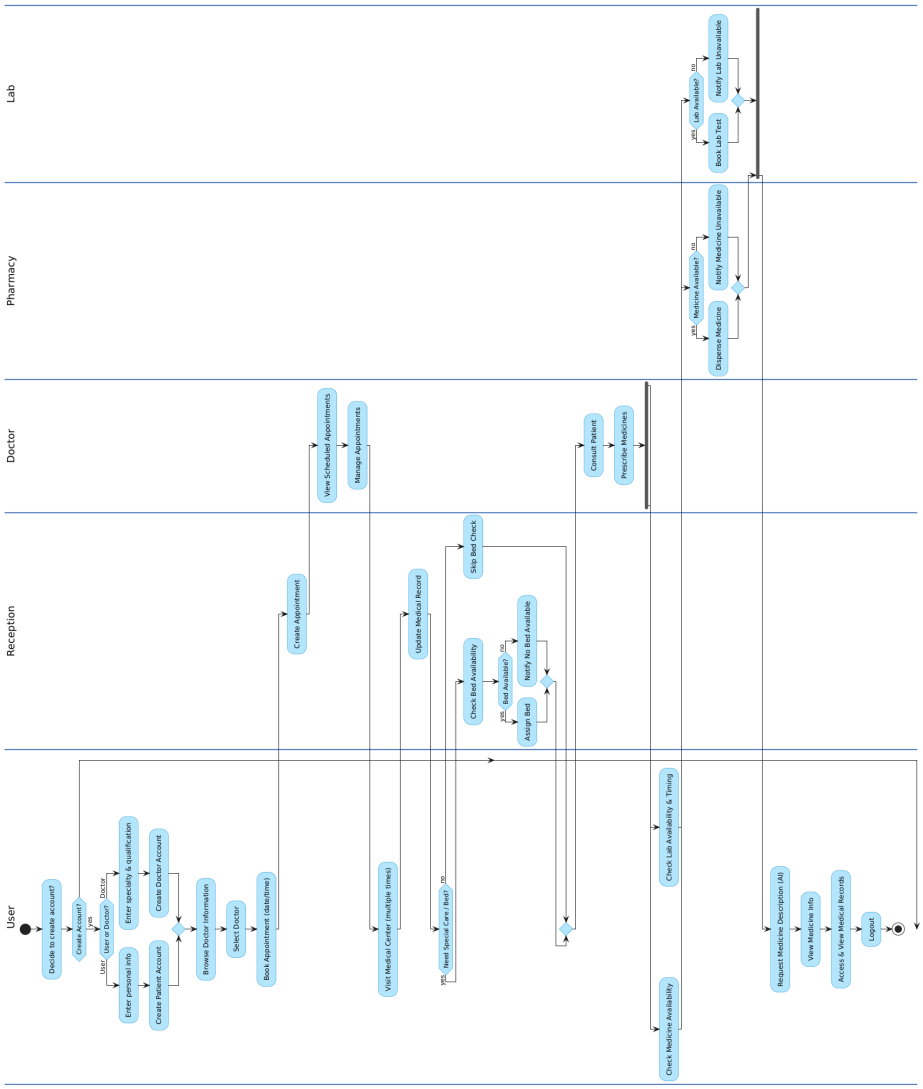

Figure 2: Activity Diagram of CUET Medical

3.3 Class Diagram

Class Diagram Description

User (Abstract Class)

- Represents the base class for all users of the system.
- Includes attributes:
  - `userID`: Unique identifier for the user.
  - `fullName`: Full name of the user.
  - `email`: Email address of the user.
  - `password`: User's password for authentication.
  - `userType`: Defines role (Patient, Doctor, Staff).

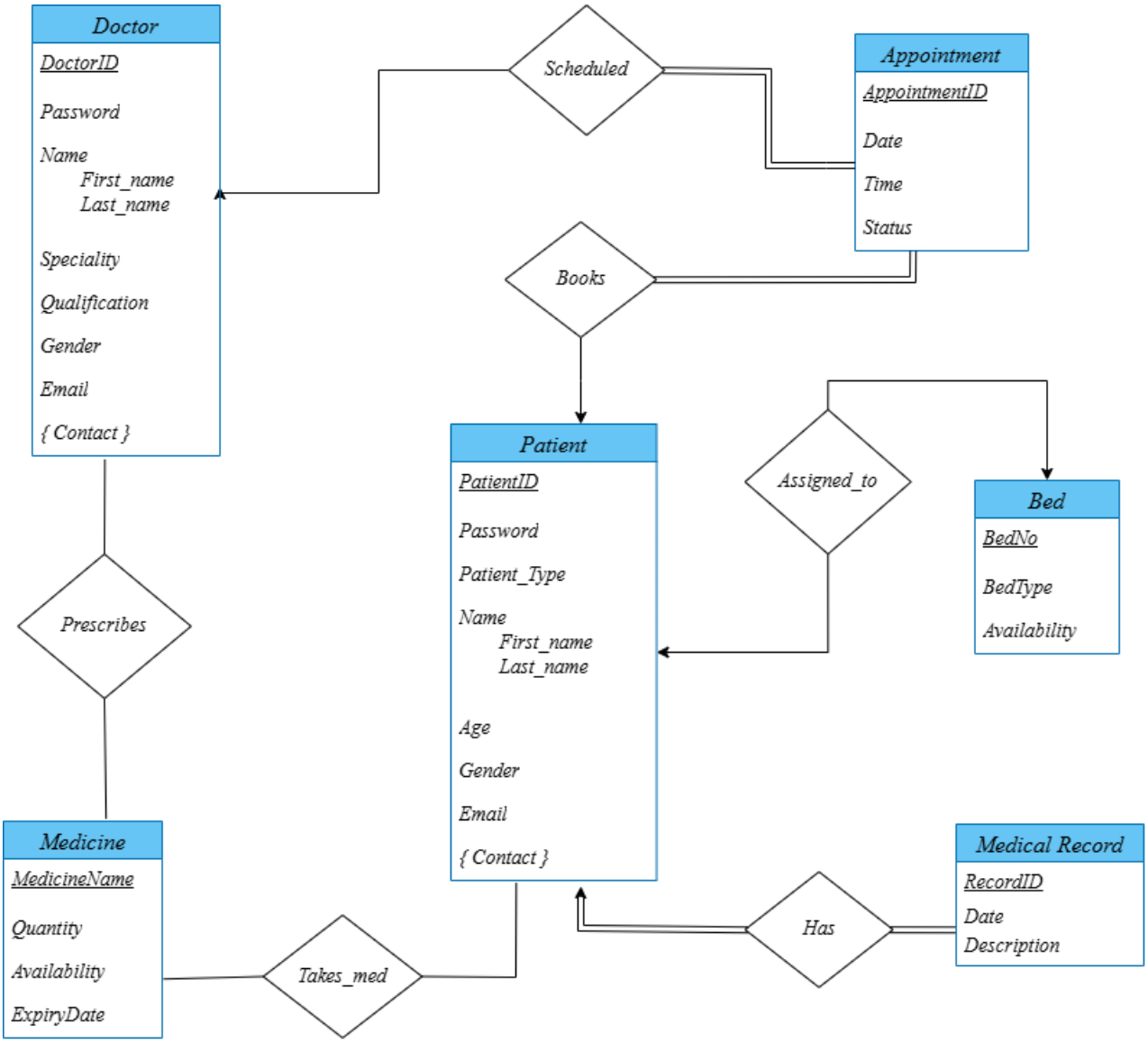

Figure 3: Class Diagram of CUET Medical

- Provides methods:
  - `login()`: Authenticates user credentials.
  - `logout()`: Ends user session.
  - `register()`: Handles new user registration.

Patient

- Inherits from User class.
- Includes attributes:
  - `patientID` (may be same as `userID`).
  - Medical history and records.
- Provides functions:
  - `bookAppointment()`: Allows patient to schedule appointments.
  - `requestBed()`: Requests bed assignment when needed.
  - `viewMedicalRecords()`: Access patient's medical records.

Doctor

- Inherits from User class.
- Includes attributes:
  - `doctorID`.
  - `specialty`.
  - `qualification`.
  - `availabilityTime`.
- Provides functions:
  - `manageAppointments()`: Views and manages appointments.
  - `prescribeMedicine()`: Adds prescriptions for patients.
  - `updateMedicalRecords()`: Updates patient records.

Staff

- Inherits from User class.
- Includes attributes:
  - `staffID`.
  - `department` or `role`.
- Provides functions:

  - `manageBeds()`: Checks and assigns beds to patients.
  - `manageMedicineInventory()`: Updates medicine stock and availability.
  - `handleLabTests()`: Schedules and manages lab tests.

Appointment

- Represents appointment details.
- Attributes:
  - `appointmentID`.
  - `date`.
  - `time`.
  - `status`.
- Linked to one Patient and one Doctor.

Bed

- Represents beds available in the medical center.
- Attributes:
  - `bedID`.
  - `bedType`.
  - `availabilityStatus`.
- Assigned to patients when occupied.

Medicine

- Represents medicines stocked in inventory.
- Attributes:
  - `medicineName`.
  - `usage`.
  - `dosage`.
  - `sideEffects`.
  - `alternatives`.
- Can be prescribed by doctors and taken by patients.

Medical Record

- Maintains medical history entries.
- Attributes:
  - `recordID`.
  - `date`.
  - `description`.
- Linked to patients and updated by doctors.

4 Core Structure

4.1 HTML

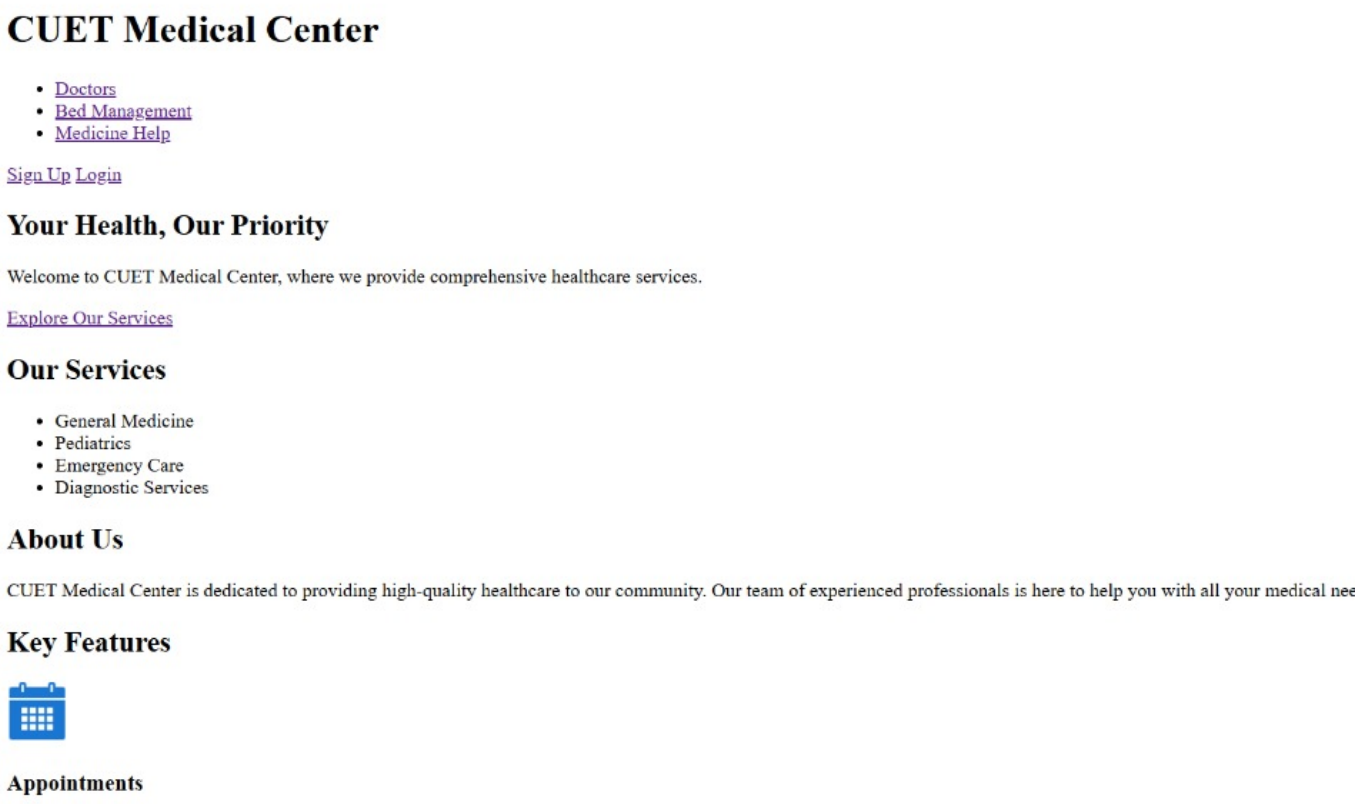

Figure 4: Landing Page

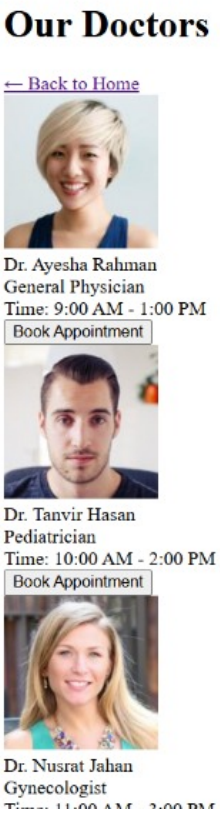

Figure 5: Available Doctors

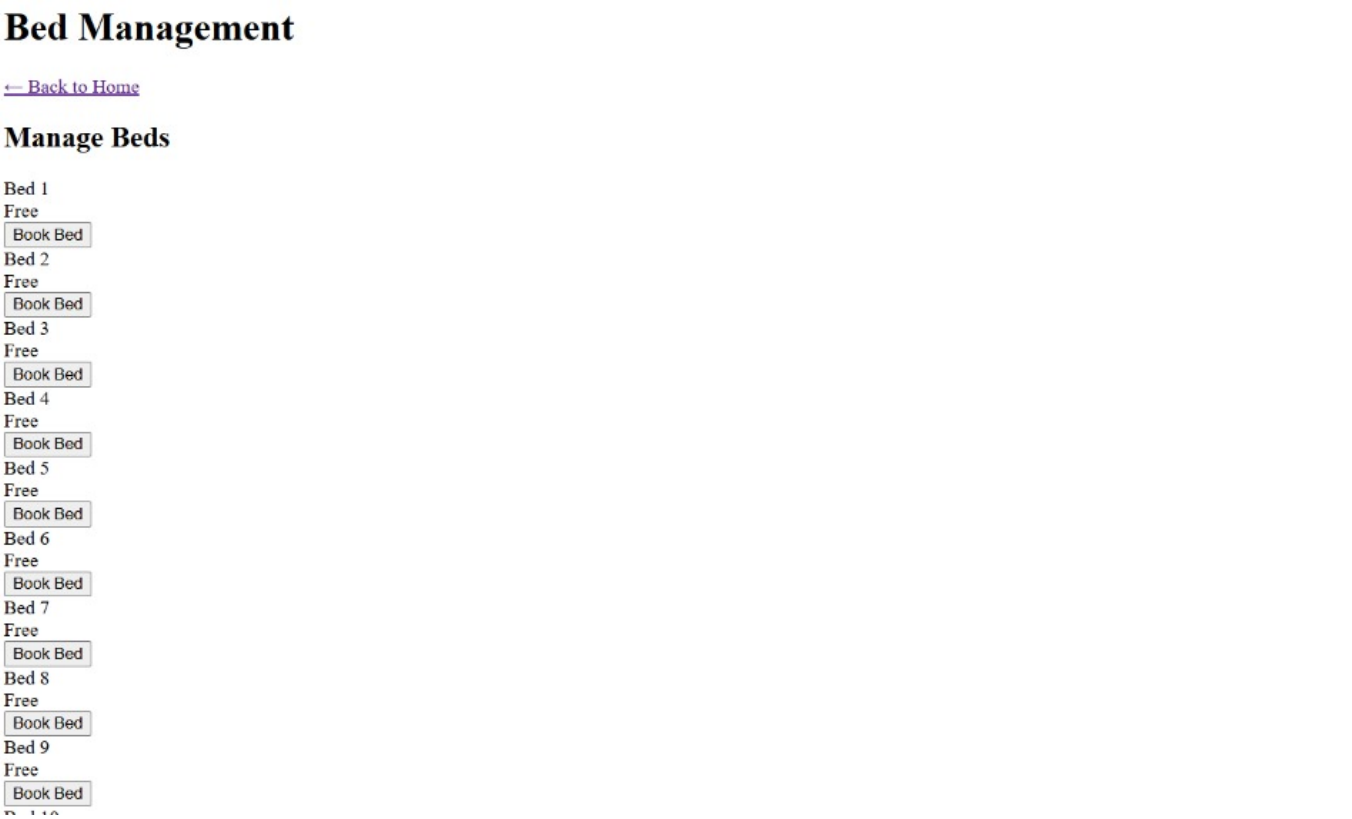

Figure 6: Bed Management

Figure 7: Medicine Inventory

5 Final Front End Design

The final design of the CUET Medical Center website is modern and user-friendly, significantly improving the basic layout. We incorporate custom colors, clean fonts, and smooth animations to create an attractive interface. Interactive elements such as doctor appointment booking, bed management system, and search for medicine information enhance usability. The intuitive navigation and responsive design ensure easy access to services. Together, these features offer users a seamless and engaging experience on the platform.

5.1 HTML with CSS

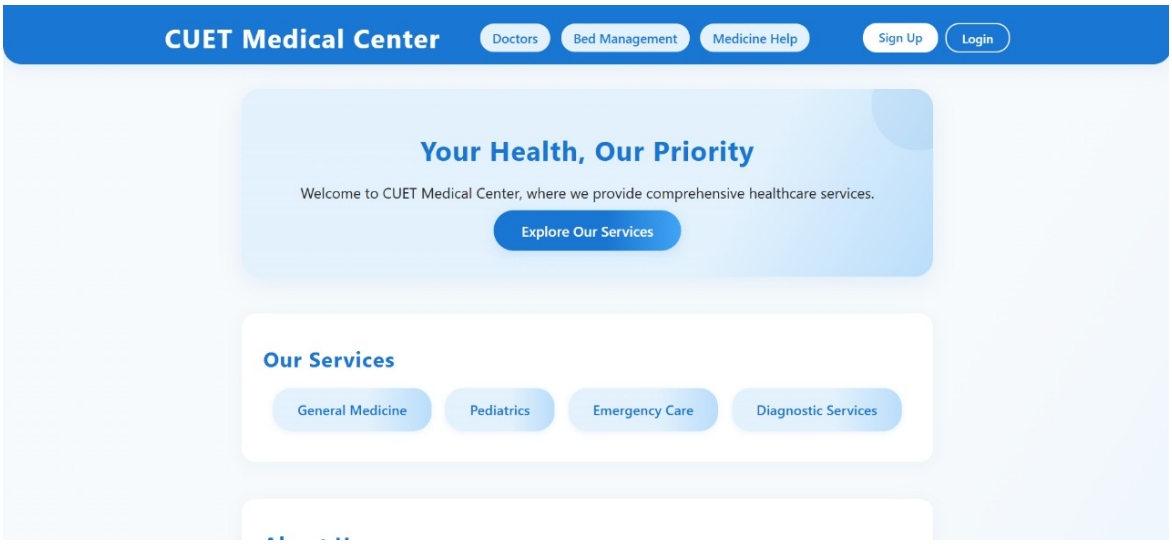

Figure 8: Landing Page

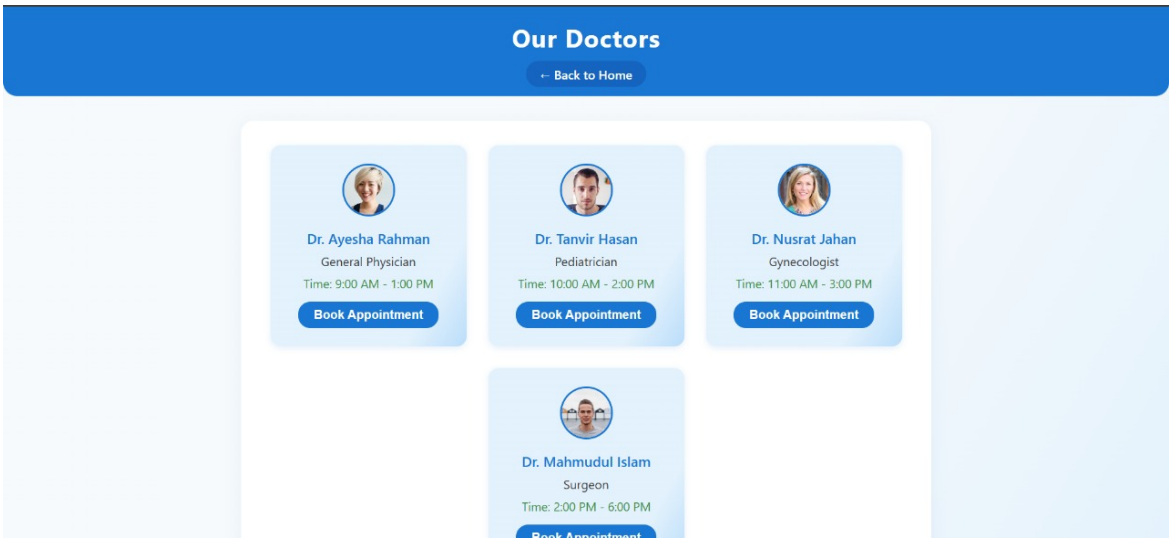

Figure 9: Available Doctors

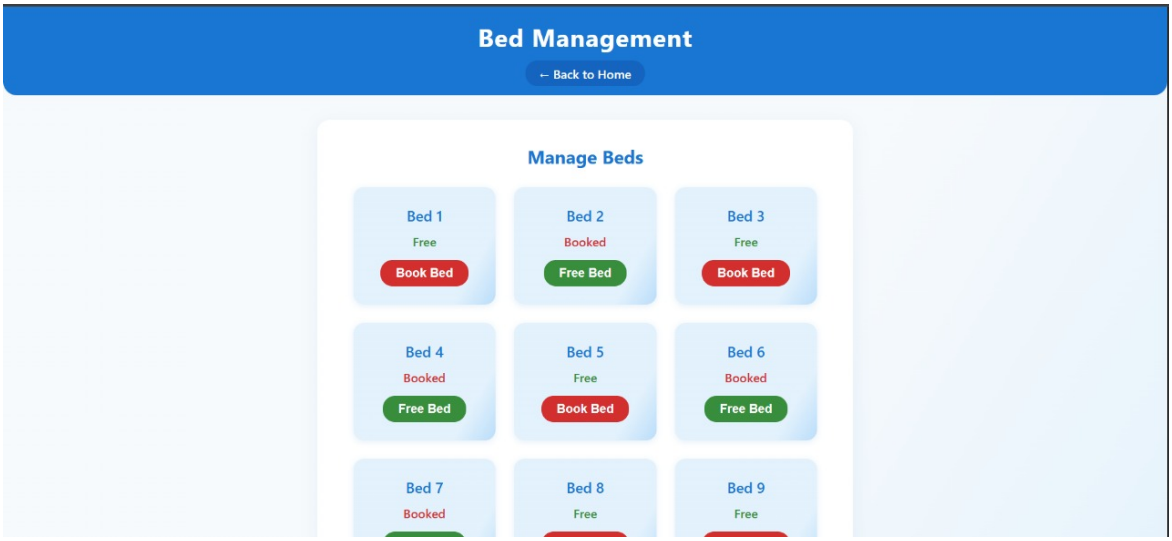

Figure 10: Bed Management

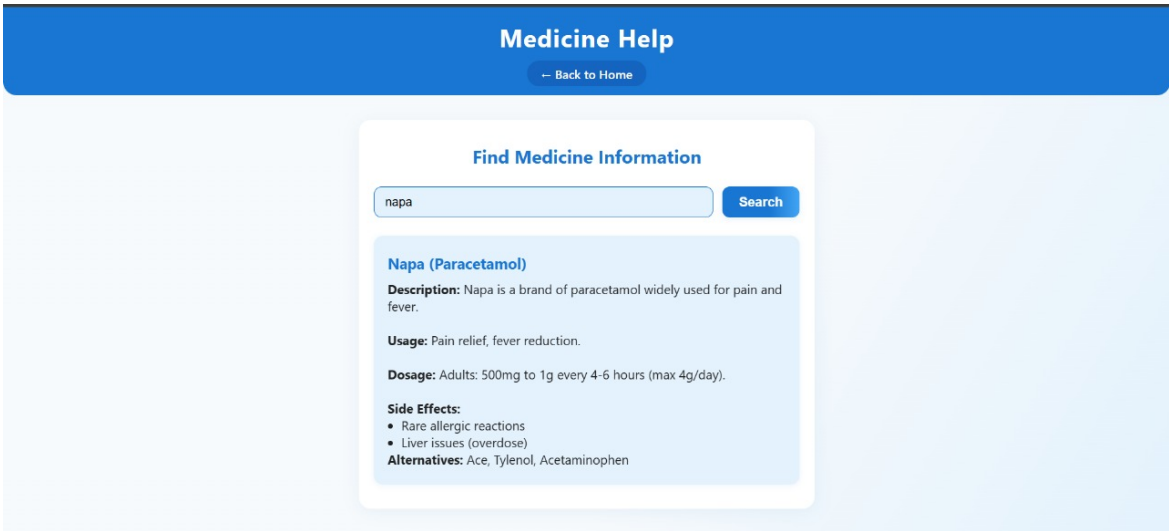

Figure 11: Medicine Inventory

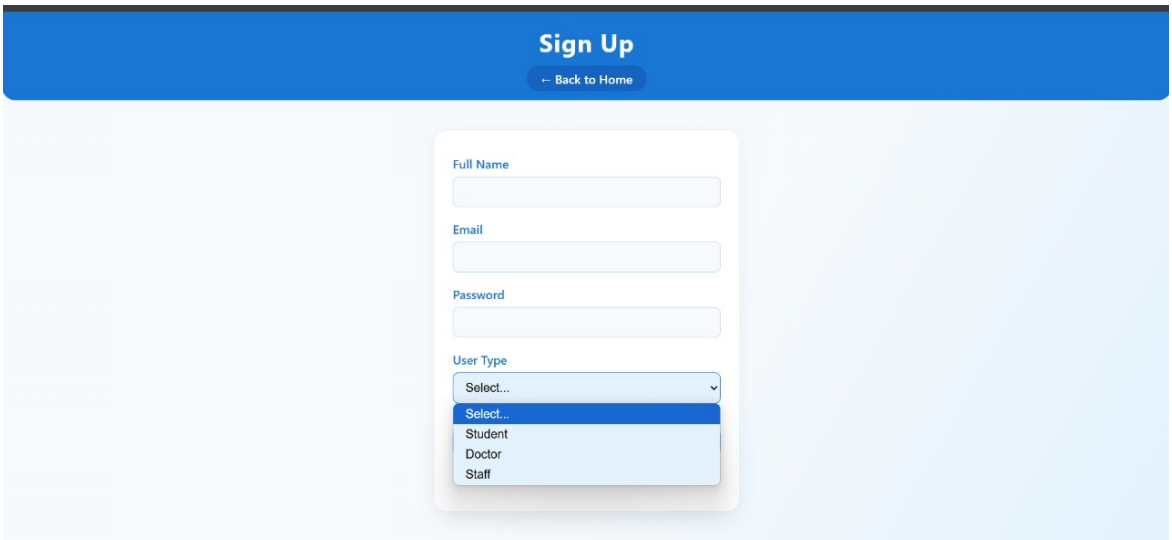

Figure 12: Sign Up Page

6 Discussion

This medical management system will develop healthcare services for students, faculty and staff. By digitizing the health record, simplifying appointment scheduling, and ensuring efficient inventory management, the system improves the operational efficiency of our medical center. It will provide easy access to patient data, which is crucial for informed decision making and continuity of care. This system will improve the overall workflow of our medical center compared to this traditional paper-based system.